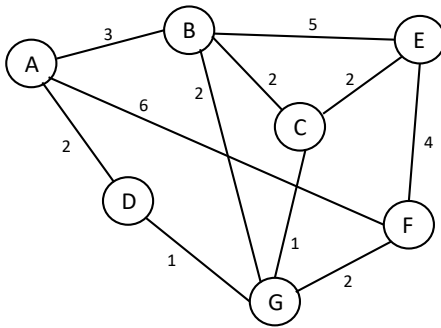


ECE-UY 3613 2024 Spring Final Exam Book

Name:

NetID:

- Write your NAME and ID on the exam book;
 - Write your answers CLEARLY and INTELLIGIBLY on the front pages only;
- 1) (12 points) Apply Dijkstra's shortest path routing algorithm to the following network to construct the shortest paths from node A to all the other nodes in the network.
- (9 points) Show the steps of Dijkstra algorithm by providing a table similar to the homework question;
 - (3 points) Show the routing table on A to all the other nodes.



N'	B	C	D	E	F	G
A	3, A	∞	<u>2, A</u>	∞	6, A	∞
AD	<u>3, A</u>	∞		∞	6, A	3, D
ADB		5, B		8, B	6, A	<u>3, D</u>
ADB G		<u>4, G</u>		8, B	5, G	
ADBG C				6, C	<u>5, G</u>	
ADBG CF				<u>6, C</u>		

Routing Table

dest.	Next hop
B	(A, B)
C	(A, D)
D	(A, D)
E	(A, D)
F	(A, D)
G	(A, D)

- 2) (8 points) In the Distance Vector (DV) shortest path algorithm,
- (3 points) if node s only has three neighbors, x, y, z . The link weights from s to x, y, z are 2, 4, 3, respectively. If the shortest distances from x, y, z to some destination d are 10, 12, 8 respectively. What is the shortest distance from s to d ?
 - (5 points) suppose all nodes run DV algorithm synchronously, after the k -th iteration, if the distance estimate from node A to some destination x is $D_A^k(x) = 10$, what is the possible value range for A 's distance estimate to x , $D_A^{k-1}(x)$, in the previous iteration? if the weight on link $\langle A, B \rangle$ is 2, what is the possible value range for B 's distance estimate to x , $D_B^{k+1}(x)$, in the next iteration?

Answer: a) the shortest path distances on s through the three neighbors x, y, z are $2 + 10 = 12$, $4 + 12 = 16$ and $3 + 8 = 11$, respectively. So the shortest path distance is 11.

b) $D_A^k(x)$ either decreases or remains the same over iterations, so $D_A^{k-1}(x) \geq D_A^k(x) = 10$;

At iteration $k + 1$, B will receive distance estimation $D_A^k(x) = 10$ from A, so the distance estimate on B through A is $D_A^k(x) + 2 = 12$, so $D_B^{k+1}(x) \leq 12$.

- 3) (8 points) Please answer the following questions on BGP:

- (4 points) Does BGP routing always select the path with the least number of ASes? Briefly explain why or why not?
- (4 points) What is the hot-potato routing? Briefly explain the motivation for it.

Answer: a) BGP is policy based routing, it does not always take the path with the least number of ASes. For example, if a neighbor AS on the shortest path does not agree to forward traffic, traffic has to be routed on a longer path.

b) hot-potato routing is that for a packet destined to another AS, a router will forward it to a border router with the least intra-AS cost that can reach the other AS. This is because each AS is interested in minimizing its own traffic cost, without considering the traffic cost in other ASes.

4) (12 points) When CRC is employed for error detection, given an original message of 1101011 and generator bit sequence of 1011,

- (8 points) what are the CRC check bits generated? what is the transmitted message with CRC bits?
- (4 points) suppose during the transmission the first bit and the last bit of the transmitted message are flipped, what is the bit sequence received by the receiver? can the receiver detect this error using CRC check? support your answer by showing the calculation.

Handwritten calculations for CRC error detection:

Generator: 1011

Original Message: 1101011

Transmitted Message: 1101011000

Received Message: 010101111 (Note: first and last bits are flipped)

Receiver Check:

1011 $\overline{) 101011111}$

1011

1111

1011

1001

1011

1000

1011

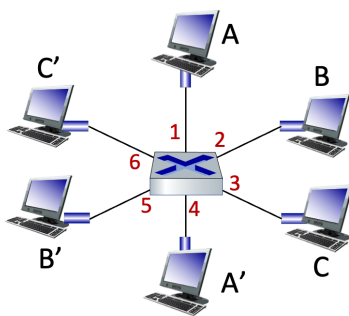
110

101 — non-zero remainder
error detected

5) (10 points) For the switched Ethernet in the following figure, suppose the switch table was initially empty, please describe the content of the switch table and the switch forward operation for the following three consecutive frames:

1) From A to B'; 2) From C to A; 3) From B to B'

You can use TTL value of 60 seconds for each new entry in the switch table.



1) $A \rightarrow B'$, flood to ports 2-6

host	port	TTL
A	1	60

2) $C \rightarrow A$, send to port 1,

host	port	TTL
A	1	60
C	3	60

→ refreshed by C→A frame

3) $B \rightarrow B'$ flood to ports 1, 3, 4, 5, 6

host	port	TTL
A	1	59
C	3	59
B	2	60

6) (10 points) Answer the following questions for medium access control:

a) (4 points) An Ethernet LAN has three stations, if all the three stations transmit at the same time and ran into a collision, what is the chance that the next transmission attempt on the LAN will be successful?

b) (6 points) What is the hidden terminal problem for wireless LAN? How is it solved?

Answer: a) after the first collision, each of the three stations will choose a random number between (0, 1) for back-off, to make sure the first attempt is successful, we need exactly one station choose 0, the other two stations both choose 1, the chance it happens is $3/8$

b) When A and B both transmit to C in the middle of them, their signals will collide on C, but A and B cannot hear each other's transmissions, because the signal gets weaker at longer distance.

The problem can be solved using CSMA/CA, where the receiver will explicitly acknowledge the receipt of a frame, and RTS/CTS: when a sender wants to transmit a frame, it first transmits a Request-to-Send frame, when the receiver receives RTS requests, it will choose one station to grant Clear-to-Send (CTS), and it will be broadcasted to all stations, only the station with CTS will transmit.

In the previous example, both A and B send RTS to C, C will choose one station, say A, to grant CTS. Both A and B will get the CTS, so that A will transmit, and B will keep silent.

- 7) (12 points) Suppose both Alice and Bob have their own public-private key pairs. They have each other's certificates, and they share the same hash function,
- (4 points) how to design a scheme for Alice to send a confidential message to Bob?
 - (4 points) if the message is too long for public-private key based encryption, how to design a scheme for Alice to authenticate herself and her message to Bob?
 - (4 points) if the message is too long for public-private key based encryption, how to design a scheme for Alice to send a confidential message to Bob?

Answer: a) Alice will encrypt the message using Bob's public key; Bob will decrypt the message using his own private key;

b) Alice will apply the hash function on the message to generate a message digest, then encrypt it using her own private key, send the encrypted digest along with the message to Bob. This can authenticate Alice and her message to Bob, because only Alice has access to her private key.

c) Alice can generate a symmetric key S , then encrypt it with Bob's public key, send it to Bob. Finally, Alice and Bob will use S for message encryption and decryption.

- 8) (8 points) In digital signature, if an attacker can inverse the hash function $H(m)$ (in other words, for any given hash value h , the attacker can find a message x such that $H(x) = h$), when Alice sends a message to Bob with her digital signature, describe a way for the attacker to change the message of Alice and Bob will not notice the change.

Answer: Suppose Alice's message is M , her digital signature is $K_A^-(H(M))$. The attacker has Alice's public key K_A^+ , she can recover the message digest as $K_A^+(K_A^-(H(M))) = H(M)$, then she can find another message M' such that $H(M') = H(M)$, she will change the message to M' , transmits it along with the digital signature of $K_A^-(H(M))$. When Bob verifies the message, he cannot identify the change, since $H(M') = H(M)$.

- 9) (8 points) A mobile phone M has a permanent address of X in its home network. When it is roamed to a visited network, it gets a temporary address of Y . A correspondent C wants to send a message to M .
- (4 points) if mobility is handled by *direct routing*, which address should C use to reach M ? if mobility is handled by *indirect routing*, which address should C use to reach M ?
 - (4 points) Please list one advantage and one disadvantage for mobility handled by direct routing and indirect routing respectively.

Answer: a) With direct routing, the message will be sent to Y ; with indirect routing, the message will be sent to X ;

b) direct routing: advantage: traffic go directly to M , avoid triangle routing; disadvantage: C needs to keep track of M 's current address when M roams to different networks.

indirect routing: advantage: C always use the same address when M roams to different networks, a connection can be maintained when M roams to different networks; disadvantage: traffic go to home network first, then forwarded to M , longer path resulted from triangle routing;

- 10) (12 points) Please answer the following synthesis questions:

- (3 points) when Alice brings her laptop to school and connects to campus WiFi network, what is the application layer protocol her laptop uses to get an IP address? what is the transport layer protocol used by that protocol? what is the destination MAC address of the first data frame sent out by that protocol?
- (3 points) how does Alice's laptop find out the IP address and MAC address of the first-hop router?
- (3 points) when Alice wants to visit her online bank account securely, which application layer and transport layer protocols are used?
- (3 points) when Alice sends out a packet to the online bank server, what are the source and destination IP addresses, as well as the source and destination MAC addresses in the data frame carrying the packet?

Answer: a) DHCP, UDP, FF-FF-FF-FF-FF-FF (broadcast MAC address)

b) DHCP reply will carry the IP address of the first-hop router, ARP will be used to find its MAC address

c) HTTP/TLS/TCP (or HTTP/QUIC/UDP)

d) Alice's IP address, server's IP address, Alice's MAC address, first-hop router's MAC address.

+++++++ End of Exam, Enjoy Your Summer! +++++++

If you use up the space under any particular problem, you can write your answer here. On the page of the problem, tell us part of your work is here and write down the page number of this page and. Don't tear off any pages.

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